



Ducks in a Row

Certificate Lifecycle Automation & Management for ADCS

Installation & Configuration Guide

Make ADCS work like Let's Encrypt for your internal network.

Version 0.9.0-beta.1

2026

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1. Quick Start

Get Ducks in a Row from a fresh download to a working ACME proxy. Plan for about fifteen minutes the first time, most of it spent on Active Directory permissions.

Step 1: Install Prerequisites

If you install with **Ducks-in-a-Row-Setup.exe** (recommended), the ASP.NET Core 8.0 runtime is installed for you, offline, from the installer itself. If you install the bare MSI instead, first download and install the **ASP.NET Core 8.0 Hosting Bundle** from the official .NET download page. Either way, install the ADCS management tools and reboot the server if prompted.

```
# Bare MSI only: download and install the .NET 8 Hosting Bundle from:
# https://dotnet.microsoft.com/download/dotnet/8.0

# Then install ADCS Remote Administration Tools (elevated PowerShell):
Install-WindowsFeature RSAT-ADCS-Mgmt
```

Critical: The ADCS RSAT tools must be installed on the Ducks in a Row server even though it is not the CA server. Ducks in a Row communicates with ADCS through the ADCS COM classes, which are registered by the RSAT feature. Without them, CA connectivity fails with COM activation errors.

Step 2: Run the Installer

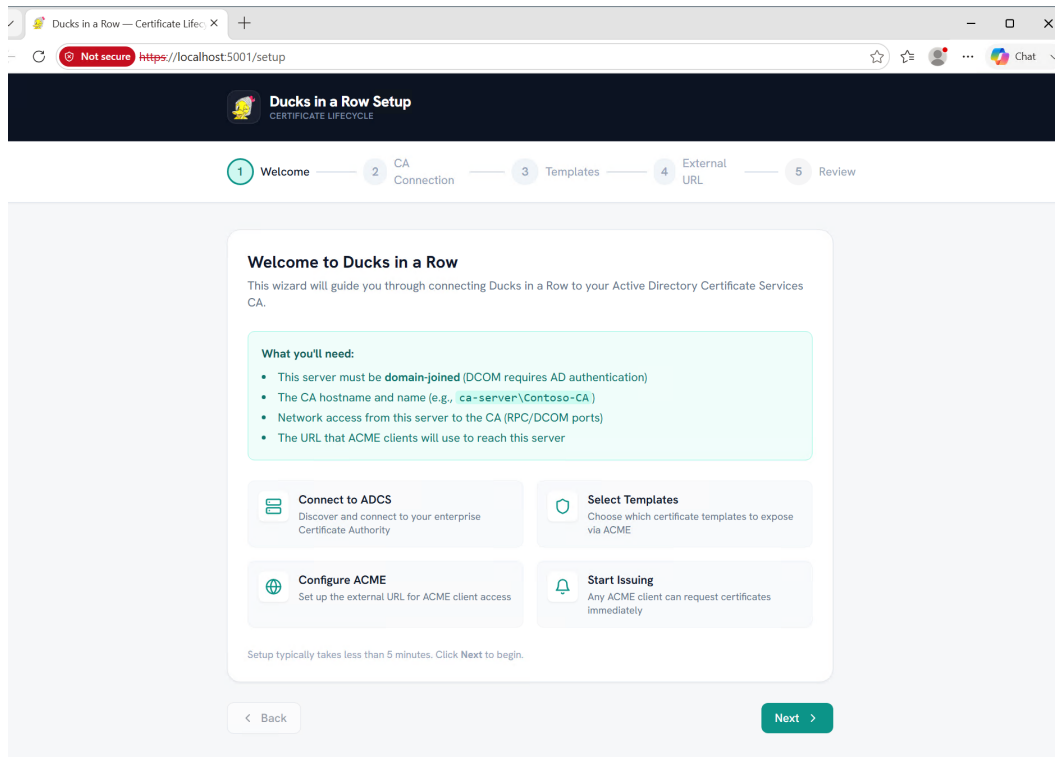
Run Ducks-in-a-Row-Setup.exe. It installs the ASP.NET Core runtime when the server does not have it, then opens the install wizard. On a server that already has the runtime, the bare MSI also works, either directly or from an elevated command prompt:

```
msiexec /i Ducks-in-a-Row.msi
```

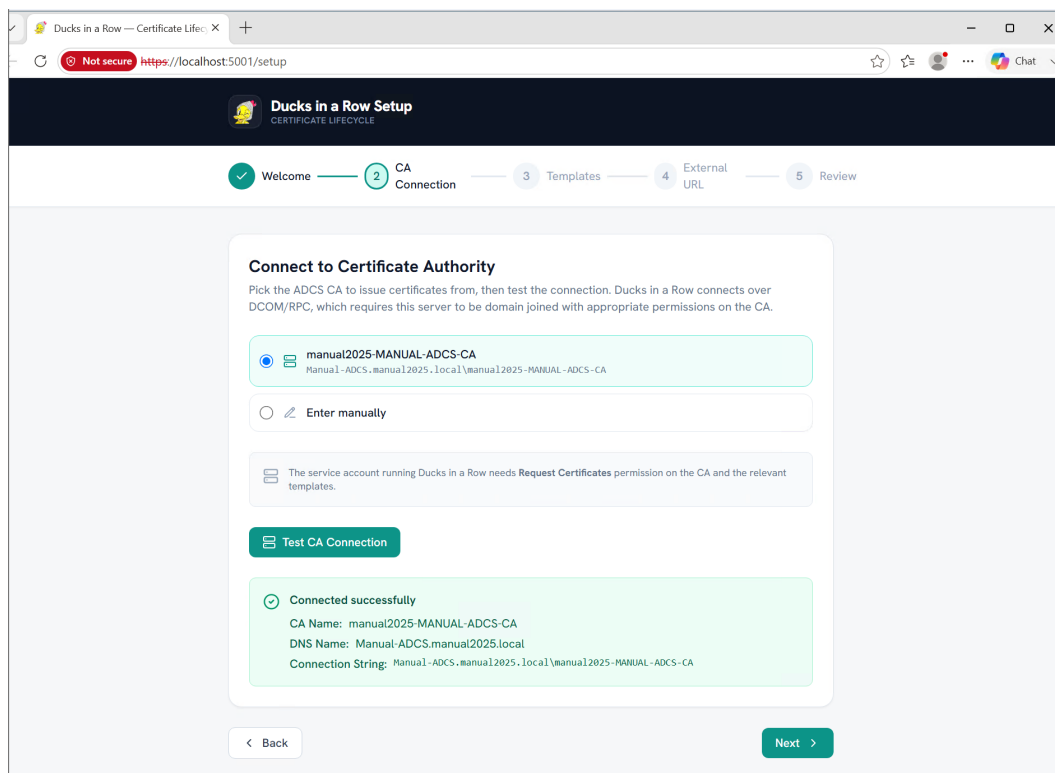
The install wizard asks for the destination folder, the data folder (database, logs, and runtime configuration), and whether to start the service immediately.

Step 3: Run the Setup Wizard

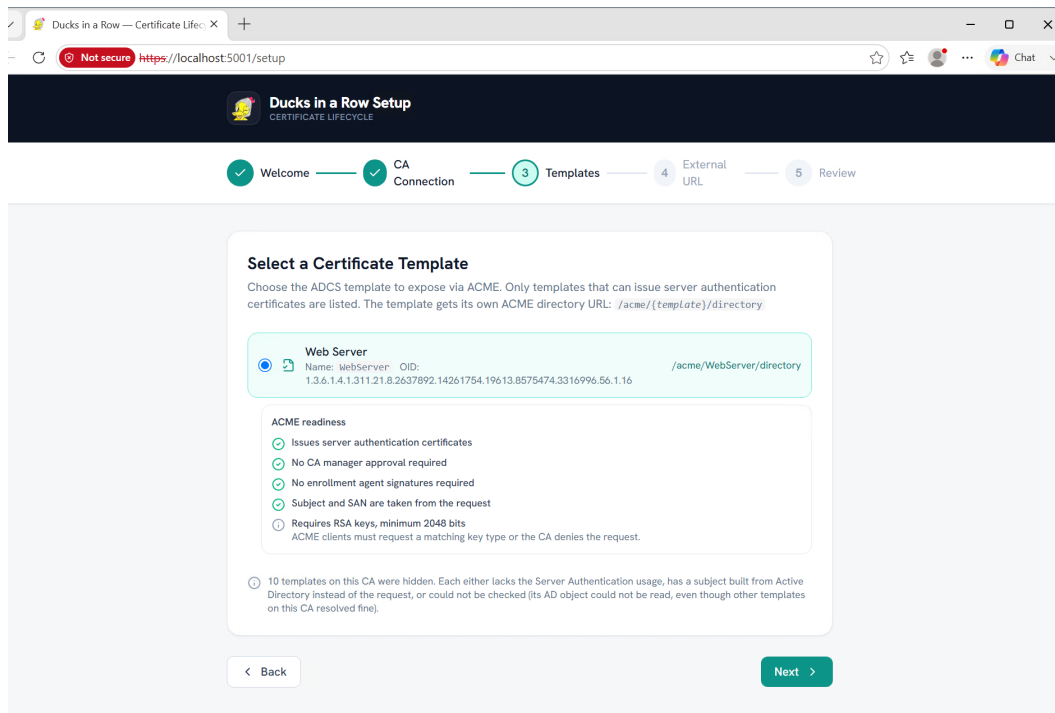
Open the setup page (the completion screen offers to open it for you) and sign in as a member of the administrator group. The wizard discovers the CAs published in Active Directory, tests the connection to the CA you pick, lists its templates and lets you choose one, validates the external URL, then applies the configuration and restarts the service by itself.



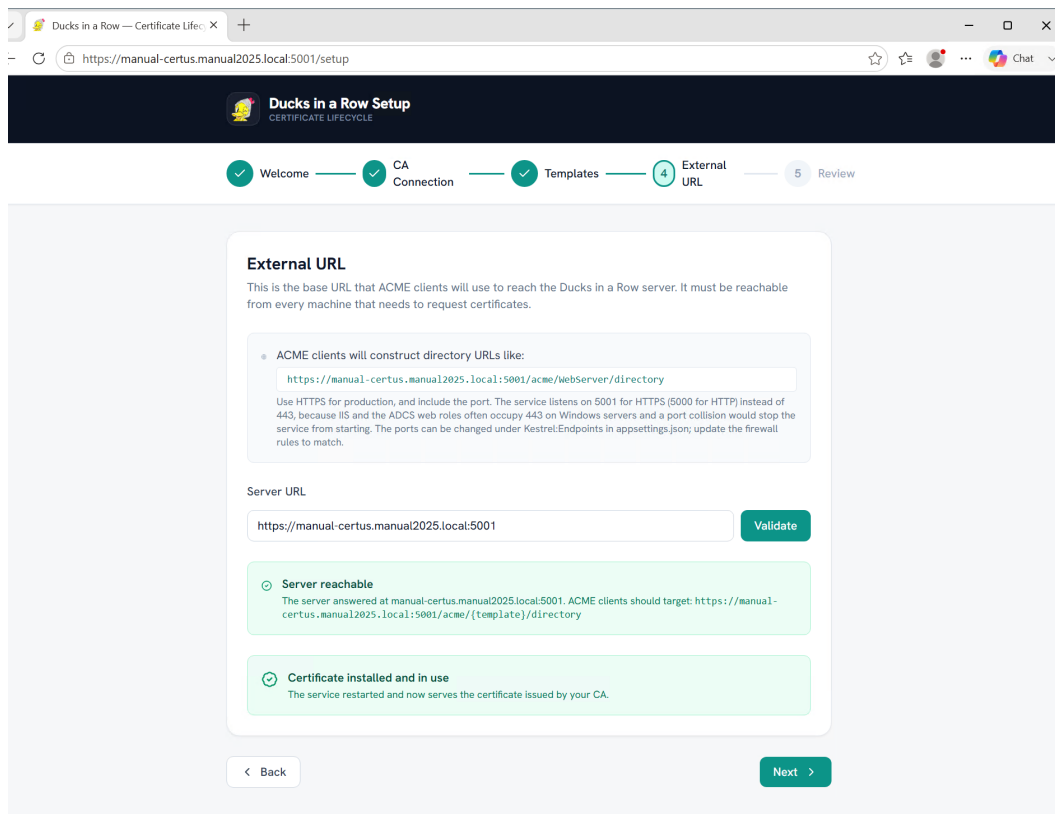
Welcome step: what you need before you start.



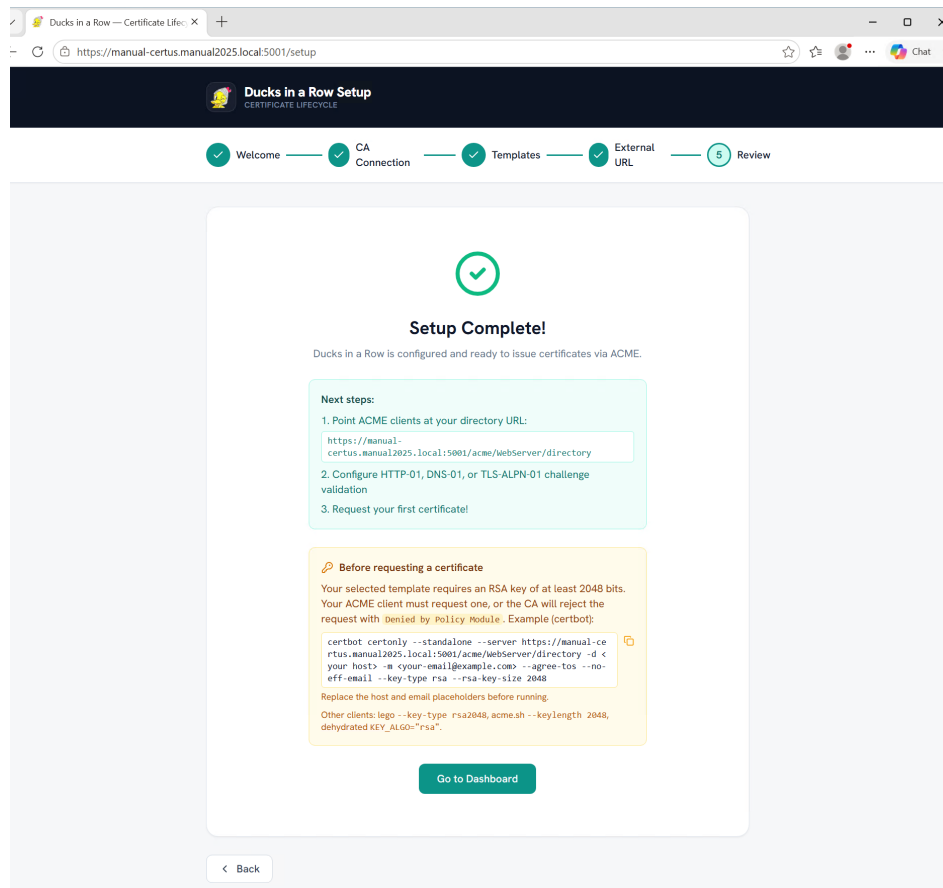
Connection step: pick a discovered CA and test it.



Templates step: choose a template and check ACME readiness.



External URL step: confirm the URL and, optionally, enrol an HTTPS certificate.



Review step: apply the configuration and copy the certbot command.

Step 4: Verify

Open a browser and navigate to the following URLs:

```
https://your-server:5001/health (should return Healthy)
https://your-server:5001/setup (setup wizard)
https://your-server:5001 (dashboard)
```

The HTTPS endpoint uses a self signed certificate out of the box, so your browser warns on first visit. Plain HTTP on port 5000 is available for lab use.

Note: Until the setup wizard completes, the service runs unconfigured: the wizard and dashboard are reachable, and CA operations answer 503.

2. System Requirements

Server Requirements

Requirement	Details
Operating System	Windows Server 2016 or later (2019, 2022, 2025)
Runtime	ASP.NET Core 8.0 (installed automatically by Ducks-in-a-Row-Setup.exe; install the Hosting Bundle manually only for the bare MSI)
Windows Feature	RSAT-ADCS-Mgmt (ADCS Remote Administration Tools)
Processor	2 x64 cores minimum. The work is mostly I/O bound (COM and RPC calls to the CA, plus SQLite), with short bursts of certificate crypto
Memory	1 GB minimum, 2 GB recommended
Disk	About 500 MB for the application and the .NET runtime. Plan at least 5 GB free for database growth, logs and upgrades. At roughly 1000 ACME accounts renewing over 3 years the database reaches the low hundreds of MB; log files are capped at 30 daily files
Network	TCP 5000 (HTTP) and TCP 5001 (HTTPS), both configurable
Domain	Must be domain-joined for ADCS connectivity

Installing Prerequisites

Run the following commands in an elevated PowerShell session on the Ducks in a Row server before installing:

```
# Install ADCS Remote Administration Tools
Install-WindowsFeature RSAT-ADCS-Mgmt

# Verify the feature is installed
Get-WindowsFeature RSAT-ADCS-Mgmt

# Verify the .NET 8 runtime is installed (after the setup bundle or
# a manual Hosting Bundle install)
dotnet --list-runtimes
```

Why RSAT-ADCS-Mgmt is required: Ducks in a Row communicates with the ADCS Certification Authority through the ADCS COM classes, which are registered on the system by the ADCS Remote Server Administration Tools. Without this feature installed, the COM objects cannot be instantiated and the service fails with `DISP_E_MEMBERNOTFOUND` (0x80020003) errors during certificate sync and CA queries.

ADCS Requirements

To issue certificates through ADCS, the following must be in place:

- An Active Directory Certificate Services (ADCS) Certification Authority (Enterprise CA)

- ADCS Remote Administration Tools (RSAT-ADCS-Mgmt) installed on the Ducks in a Row server
- DCOM/RPC network access from the Ducks in a Row server to the CA server (TCP 135 + dynamic ports 49152-65535)
- The server's machine account must have Enroll permission on the target certificate templates
- The server's machine account must have Read permission on the CA itself, or the dashboard inventory stays empty
- Certificate templates must be published to Active Directory

Network Ports

The following network ports must be open between the Ducks in a Row server and other systems:

Direction	Port	Protocol	Purpose
Inbound to the server	5000, 5001	TCP	ACME clients and dashboard browsers (HTTP and HTTPS)
Outbound to CA	135	TCP	DCOM/RPC endpoint mapper
Outbound to CA	49152-65535	TCP	DCOM/RPC dynamic ports
Outbound to targets	80	TCP	HTTP-01 challenge validation
Outbound to DNS	53	TCP/UDP	DNS-01 challenge validation
Outbound to targets	443	TCP	TLS-ALPN-01 challenge validation

3. Installation

What the Installer Does

The Ducks in a Row MSI installer performs the following actions automatically:

Action	Details
Install application files	C:\Program Files\Ducks in a Row\ (or the folder you choose)
Create the data folder	C:\ProgramData\Ducks in a Row\ (or the folder you choose; recorded in the registry)
Create logs directory	logs\ inside the data folder
Register Windows Service	DucksInARow (auto-start, runs as LocalSystem)
Start the service	Service starts immediately after install (optional)
Create firewall rules	Inbound TCP 5000 and 5001 (Ducks in a Row Certificate Proxy)
Add a Start Menu shortcut	Ducks in a Row shortcut that opens the web UI

Silent Installation

For automated deployments, use a silent install with logging. INSTALLFOLDER, DATAFOLDER, and START_SERVICE are optional properties:

```
msiexec /i Ducks-in-a-Row.msi /qn INSTALLFOLDER="D:\Ducks in a Row" DATAFOLDER="D:\DucksData" START_SERVICE=0 /l*v "C:\temp\ducks-install.log"
```

INSTALLFOLDER and DATAFOLDER default to the Program Files and ProgramData folders shown above. Choose the data folder at install time; moving it on a later upgrade is not supported. START_SERVICE=0 registers the service but leaves it stopped. A /qn install never launches a browser.

Service Recovery

The Ducks in a Row service is configured to restart automatically on failure. The recovery policy uses escalating delays:

Failure	Action	Delay
First failure	Restart the service	5 seconds
Second failure	Restart the service	10 seconds
Subsequent failures	Restart the service	30 seconds

4. Configuration

Configuration is layered. Shipped defaults live in `appsettings.json` in the installation folder. The setup wizard writes instance settings (the CA connection string and external URL) to `settings.json` in the data folder, which overrides the defaults and survives upgrades. Environment variables override both. After editing either file, restart the service for changes to take effect.

Core Settings

Setting	Default	Description
CaConnectionString	null	ADCS CA connection string in hostname\CAName format; the setup wizard sets it
DatabasePath	ducks.db in the data folder	Full path to the SQLite database file
ExternalUrl	null	Public URL that ACME clients use to reach the server; the setup wizard sets it
SyncIntervalMinutes	5	How often to sync certificates from the ADCS CA database
EnableWalMode	true	Enable SQLite WAL mode for better concurrent performance

Authentication Settings

The dashboard and setup API use Windows Integrated Authentication and are limited to the administrator group.

Setting	Default	Description
Auth:Mode	Negotiate	Windows Integrated Authentication for the dashboard
Auth:AdminGroup	null	Group allowed into the dashboard and setup; null means the built in Administrators group
Auth:RequireHttps	true	HSTS and HTTP to HTTPS redirect outside development
Auth:TrustedProxies	[]	Reverse proxy IPs whose X-Forwarded-* headers are trusted

Minimum Required Configuration

The setup wizard writes everything the service needs. For reference, after setup the `settings.json` file in the data folder looks like this:

```
{
  "Certus": {
    "CaConnectionString": "CA-SERVER\\MyCA",
    "ExternalUrl": "https://ducks.yourdomain.local:5001"
  }
}
```

Important: The `ExternalUrl` must be reachable by ACME clients on your network. This URL is embedded in ACME directory responses and challenge URLs. Use HTTPS in production environments.

Kestrel Web Server

By default, Ducks in a Row listens on all interfaces on TCP 5000 (HTTP) and TCP 5001 (HTTPS). The HTTPS endpoint uses a self signed certificate out of the box, generated in the data folder as `ducks-selfsigned.pfx`, so browsers warn on first visit and ACME clients will not trust it.

The easiest fix is built in. On the external URL step, the setup wizard can enrol an HTTPS certificate for this server from the CA you connected, install it, and reload the service. The Settings page can renew it later. Use that first.

If you would rather supply your own certificate, point Kestrel at one your clients already trust (for example, a certificate issued by your own CA whose root your clients trust):

```
"Kestrel": {
  "Endpoints": {
    "Https": {
      "Url": "https://0.0.0.0:5001",
      "Certificate": {
        "Path": "C:\\ProgramData\\Ducks in a Row\\ducks.pfx",
        "Password": "your-pfx-password"
      }
    }
  }
}
```

For a quick lab test only, you can set `Auth:RequireHttps` to `false` and use plain HTTP on port 5000 instead.

Logging

Ducks in a Row uses Serilog for structured logging. Logs are written to `logs\` in the data folder with daily rotation and a 30-day retention policy. Adjust the minimum log level in the Serilog configuration section:

```
"Serilog": {
  "MinimumLevel": {
    "Default": "Information",
    "Override": {
      "Microsoft.AspNetCore": "Warning",

```

```
    "Microsoft.EntityFrameworkCore": "Warning"  
  }  
}  
}
```

5. ADCS Configuration

Finding Your CA Connection String

The CA connection string follows the format `hostname\CAName`. The setup wizard finds it for you, so you rarely need to build it by hand. The manual options below are a fallback.

Option A: Let the Setup Wizard Discover It

Open `https://your-server:5001/setup` after installation. The wizard discovers the ADCS CAs published in Active Directory, lets you pick one, and tests connectivity through the same path the service uses. This is the recommended way and needs no connection string typed by hand.

Option B: From the CA Server

Open the Certification Authority MMC snap-in (`certsrv.msc`) on your CA server. The CA name appears in the console tree. The connection string is the CA server hostname followed by a backslash and the CA name.

Option C: Using PowerShell

Run this on any domain-joined machine to discover CAs in your forest:

```
certutil -config - -ping
```

This displays a list of available CAs. The output includes the connection string in the required format.

Certificate Template Permissions

For Ducks in a Row to issue certificates on behalf of ACME clients, the server's machine account needs Enroll permission on each certificate template you plan to use. To configure this:

1. Open the Certification Authority MMC (`certsrv.msc`) on your CA server.
2. Right-click the target certificate template and select Properties.
3. Go to the Security tab.
4. Add the Ducks in a Row server machine account (e.g., `DUCKS-SERVER$`).
5. Grant the Enroll permission. Do not grant Autoenroll.
6. Click OK and repeat for each template.

CA Read Permission

The machine account also needs Read permission on the CA itself (Certification Authority console, CA properties, Security tab). Read is what lets the dashboard sync the certificate inventory. Without it, enrolment still works but the dashboard stays empty.

Template URL Mapping

Each ADCS certificate template is exposed as its own ACME directory endpoint. ACME clients target the URL for the specific template they need:

Template Name	ACME Directory URL
WebServer	<code>https://your-server:5001/acme/WebServer/directory</code>
(your template name)	<code>https://your-server:5001/acme/{template-name}/directory</code>

Note: The template segment is the template's programmatic name (its AD cn) or its display name, matched case insensitively. Display names with spaces work; the client URL encodes them.

6. Testing Your Installation

Service Status

Verify the Ducks in a Row service is running:

```
Get-Service DucksInARow

# Expected output:
# Status Name DisplayName
# -----
# Running DucksInARow Ducks in a Row Certificate Proxy
```

Health Endpoint

The health check endpoint reports the status of the database and CA connectivity:

```
Invoke-RestMethod https://your-server:5001/health

# Returns: Healthy, Degraded, or Unhealthy
```

Status	Meaning
Healthy	Database is accessible and CA connection is working
Degraded	Database is working but CA is not reachable
Unhealthy	Database is not accessible

ACME Directory

Test that ACME endpoints are responding correctly:

```
Invoke-RestMethod https://your-server:5001/acme/WebServer/directory | ConvertTo-Json

# Expected: JSON with newNonce, newAccount, newOrder URLs
```

Firewall Verification

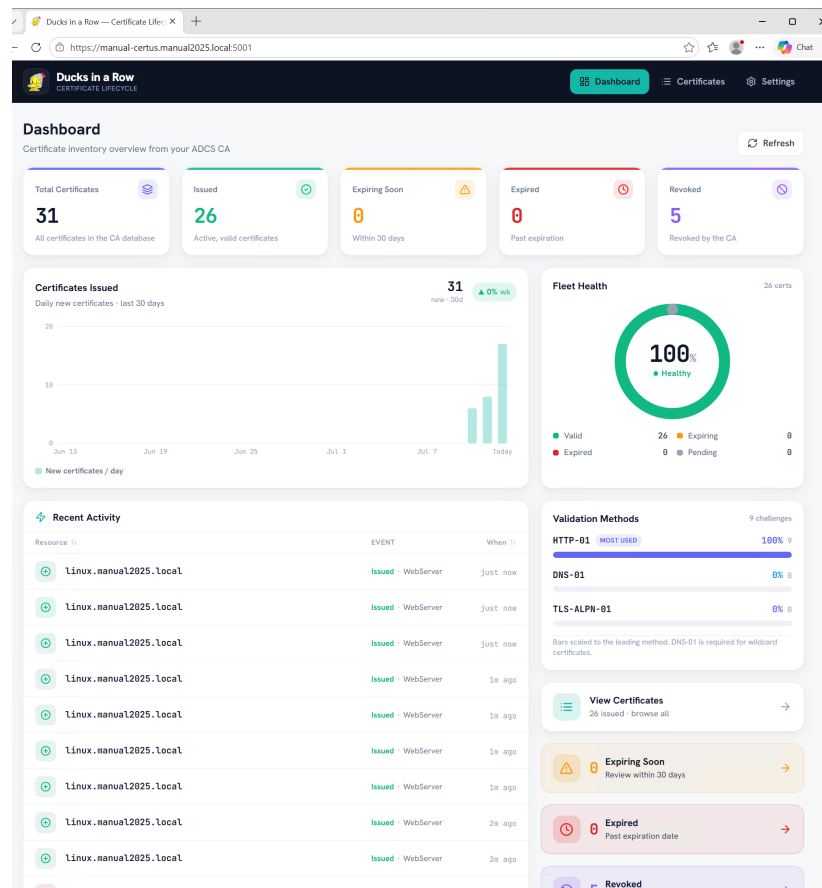
Confirm the firewall rules (HTTP and HTTPS) were created by the installer:

```
Get-NetFirewallRule -DisplayName "Ducks in a Row Certificate Proxy"
```

Dashboard Access

Open <https://your-server:5001> in a browser and sign in as a member of the administrator group. The dashboard displays certificate inventory, expiry status, and template information. On first access, you may be

redirected to the setup wizard at /setup.



The dashboard after the first sync, showing the certificate inventory and expiry tiles.

CA Connectivity Test

The setup wizard at <https://your-server:5001/setup> includes a built in connectivity test. Click **Test Connection** to verify DCOM/RPC communication with your AD CS CA. The test will report the CA name, server, and available templates.

7. Connecting ACME Clients

Ducks in a Row implements RFC 8555 (ACME), so any standard ACME client works out of the box. No agent or proprietary software is needed on your endpoints. Use the HTTPS endpoint (5001) in production; plain HTTP (5000) is for lab use only. The example commands are representative; exact flags vary by client version, so check each client's own documentation for your release.

Trusting the server TLS: ACME clients reject an untrusted TLS certificate on the ACME server itself. Give Ducks in a Row a certificate your clients already trust (for example, one chained to your internal CA root), or add that root to each client's trust store. The examples below note the per client switch for a lab where the server still uses its self signed certificate.

Certbot (Linux)

```
# HTTP-01, certbot answers the challenge itself on port 80
certbot certonly --standalone \
  --server https://your-server:5001/acme/WebServer/directory \
  --email you@example.com \
  -d host.corp.example.com \
  --key-type rsa --rsa-key-size 2048
```

Key type: If your ADCS template uses an RSA CSP (the default Web Server ACME template does), your ACME client must request an RSA key. Most modern clients default to elliptic curve keys, which the CA policy module rejects at finalize with Denied by Policy Module. The last line above forces RSA for certbot. Other clients: `lego --key-type rsa2048`, `acme.sh --keylength 2048`, `dehydrated KEY_ALGO="rsa"`.

Accounts do not need to be registered in advance and no external account binding is required; the client creates an account from its own key on first use. For a lab server with an untrusted TLS certificate, set REQUESTS_CA_BUNDLE to your CA root PEM so certbot trusts the endpoint.

win-acme (Windows)

Run wacs.exe and, in the menu, set the ACME server to the template directory URL, then create a certificate. Unattended example:

```
wacs.exe --source iis --siteid 1 ^
  --baseuri https://your-server:5001/acme/WebServer/directory
```

win-acme uses the Windows certificate trust store, so import your CA root on the machine running it if the server certificate is not already trusted.

Caddy

```
# Caddyfile
{
  acme_ca https://your-server:5001/acme/WebServer/directory
  # For an internal CA, point Caddy at the root so it trusts the ACME endpoint
  acme_ca_root /etc/ssl/certs/corp-root.pem
```

```
    email you@example.com
  }

  host.corp.example.com {
    respond "Hello from Caddy"
  }
```

Traefik

```
# traefik.yml
certificatesResolvers:
  ducks:
    acme:
      caServer: https://your-server:5001/acme/WebServer/directory
      email: you@example.com
      storage: /etc/traefik/acme.json
      httpChallenge:
        entryPoint: web
```

Traefik uses the host or container trust store. Add your CA root there if the ACME endpoint certificate is not already trusted.

Posh-ACME (PowerShell)

```
Import-Module Posh-ACME

# Point Posh-ACME at the template directory.
# Add -SkipCertificateCheck only for a lab server with untrusted TLS.
Set-PAServer -DirectoryUrl https://your-server:5001/acme/WebServer/directory

New-PAAccount -Contact you@example.com -AcceptTOS

New-PACertificate -Domain host.corp.example.com -Plugin WebRoot `
  -PluginArgs @{ WebRootPath = 'C:\inetpub\wwwroot' }
```

Challenge Types

Type	Use Case	How It Works
HTTP-01	Standard web servers	Fetches <code>http://{domain}/.well-known/acme-challenge/{token}</code>
DNS-01	Wildcards, hosts with no inbound HTTP	Looks up the TXT record at <code>_acme-challenge.{domain}</code>
TLS-ALPN-01	TLS only environments	Opens a TLS connection on port 443 and checks the ALPN certificate

The Ducks in a Row server is the party that performs validation, so the server needs outbound reachability to the domain or DNS being validated.

8. Alert Configuration

Ducks in a Row monitors certificate expiry and sends notifications through email and webhooks. Alerts are included in the free tier with no limits.

Alert Thresholds

By default, alerts fire at 30, 14, 7, and 1 day(s) before certificate expiry. Each certificate receives at most one alert per threshold (no duplicates). The monitor checks for expiring certificates every 60 minutes by default (CheckIntervalMinutes).

SMTP Email Alerts

```
"Certus:Alerts": {
  "Enabled": true,
  "CheckIntervalMinutes": 60,
  "ThresholdDays": [30, 14, 7, 1],
  "Smtp": {
    "Host": "smtp.yourdomain.local",
    "Port": 587,
    "UseSsl": true,
    "Username": "ducks@yourdomain.local",
    "Password": "your-smtp-password",
    "FromAddress": "ducks@yourdomain.local",
    "Recipients": ["admin@yourdomain.local"]
  }
}
```

Webhook Alerts

Webhooks send a JSON payload via HTTP POST. The payload is signed with HMAC-SHA256 using the configured secret. The signature is sent in the X-Certus-Signature header.

```
"Webhook": {
  "Url": "https://hooks.slack.com/services/YOUR/WEBHOOK/URL",
  "Secret": "your-hmac-secret",
  "Headers": {
    "X-Custom-Header": "value"
  }
}
```

Alert History

Sent alerts are recorded and available through the API at GET /api/alerts/history and GET /api/alerts/summary. This beta does not yet include a dedicated Alerts page in the dashboard.

9. Troubleshooting

Service fails to start

Check the Windows Event Log (Application) and the log files at `logs\` in the data folder. Common causes: missing .NET 8 runtime, invalid `appsettings.json` or `settings.json` syntax (trailing commas, missing quotes), or ports 5000 or 5001 already in use by another process. To see the actual error, run the executable directly from an elevated PowerShell: & "C:\Program Files\Ducks in a Row\DucksInARow.Service.exe"

CA operations answer 503, or the dashboard routes to the setup wizard

No CA is configured yet. The service starts unconfigured and never falls back to a fake CA on its own. Complete the setup wizard; it discovers the CAs in Active Directory, tests the connection, applies the configuration, and restarts the service.

COM activation or `DISP_E_MEMBERNOTFOUND` errors during CA operations

The ADCS COM classes are not registered on the Ducks in a Row server. Install the ADCS Remote Administration Tools: `Install-WindowsFeature RSAT-ADCS-Mgmt` and then restart the service. This feature must be installed even though this server is not the CA server. The COM class registrations are provided by the RSAT feature package.

Health endpoint returns Degraded

The database is working but the CA is not reachable. Verify the `CaConnectionString` (in `settings.json` in the data folder) is correct and uses the format `hostname\CAName`. Confirm DCOM/RPC traffic is not blocked by a firewall (TCP 135 and dynamic ports 49152-65535), and the server has network access to the CA. Test with: `certutil -config "SERVER\CA" -ping`

Health endpoint returns Unhealthy

The SQLite database is not accessible. Verify the `DatabasePath` points to a valid, writable location. Check that the data folder exists and the `LocalSystem` account has write access.

ACME client gets connection refused

Verify the service is running (`Get-Service DucksInARow`), the firewall rule exists (`Get-NetFirewallRule -DisplayName "Ducks in a Row"`), and the `ExternalUrl` in configuration matches how clients reach the server.

Certificate request fails with template error

The template name in the directory URL must match a template the CA publishes, either by its programmatic name (AD cn) or its display name; names are matched case insensitively. Confirm the template is published to Active Directory and the server's machine account has Enroll permission.

ACME client reports a TLS or certificate trust error

The HTTPS endpoint uses a self signed certificate by default (`ducks-selfsigned.pfx` in the data folder), and ACME clients reject untrusted TLS on the ACME server. Give Ducks in a Row a certificate your clients trust, or add that certificate root to each client trust store.

Cannot reach the dashboard, or get a 401

The dashboard and setup API use Windows Integrated Authentication and are limited to the administrator group. Sign in as a member of the built in Administrators group, or set `Auth:AdminGroup` to the Windows or Active Directory group you want to allow, then restart the service.

DCOM authentication errors

The Ducks in a Row server must be domain joined. Verify the machine account has not been disabled or moved to a restricted OU. Check that DCOM port ranges (TCP 135 + dynamic ports 49152-65535) are open between this server and the CA server.

Service exits silently with no log output

Run the executable directly to see the error: `dotnet "C:\Program Files\Ducks in a Row\DucksInARow.Service.dll"` Common causes include malformed JSON in `appsettings.json` or `settings.json` (use a JSON validator to check syntax), missing .NET runtime, or file permission issues. Set the environment variable `ASPNETCORE_ENVIRONMENT=Development` before running to enable detailed error output.

Dashboard shows no certificates

The machine account needs Read permission on the CA (Certification Authority console, CA properties, Security tab); without it, enrolment still works but the inventory sync cannot read the CA database and the log reports "CA view access denied". The sync also runs only every `SyncIntervalMinutes` (default 5), so wait for the first cycle or restart the service to force one.

10. Uninstalling

Ducks in a Row can be removed through Windows Add/Remove Programs or from the command line:

```
msiexec /x Ducks-in-a-Row.msi
```

What the uninstaller removes:

- All application files from C:\Program Files\Ducks in a Row\
- The DucksInARow Windows Service registration (the service is stopped first)
- The Ducks in a Row Certificate Proxy firewall rules

What the uninstaller preserves:

- The SQLite database (ducks.db in the data folder)
- Log files (logs\ in the data folder)
- The data folder itself (default C:\ProgramData\Ducks in a Row\)

To perform a complete removal including all data, delete the data directory manually after uninstalling:

```
Remove-Item -Recurse -Force "C:\ProgramData\Ducks in a Row"
```